

CHAPTER – 2

SHOPSY ONLINE E-COMMERCE DATABASE MANAGEMENT SYSTEM

2.0 Introduction :

The entities required for the **Shopsy Online E-Commerce Database Management System** are identified based on the business requirements of an online shopping platform. The attributes of each entity, along with their Primary Keys, Foreign Keys, Unique Keys, and attribute types, are determined to organize the data efficiently. The relationships between entities represent the interactions within the system and help in designing a well-structured relational database. The cardinality and participation constraints are also identified to define how entities are connected. This analysis serves as the foundation for designing the Entity Relationship (ER) Diagram, Relational Schema, and database implementation while ensuring data integrity, consistency, and minimum redundancy.

2.1 Entity Analysis :

Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
Customer_Name	Not Null
Email	Unique Key
Phone	-
Address	-
Date_of_Birth	-

Entity: Seller

Attribute	Key Type
Seller_ID	Primary Key

Attribute	Key Type
Seller_Name	Not Null
Email	Unique Key
Phone	-
Address	-

Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Seller_ID	Foreign Key
Category_ID	Foreign Key
Price	-
Stock	-
Description	-

Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key

Attribute	Key Type
Customer_ID	Foreign Key
Total_Items	-
Grand_Total	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Total_Amount	-

Entity: Order_Item

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Entity: Inventory

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Available_Stock	-
Last_Updated	-

Entity: Delivery

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	-
Delivery_Date	-
Delivery_Status	-

Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

Entity Relationships :

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Seller	Sells	Product	One-to-Many	Total / Total
Category	Categorizes	Product	One-to-Many	Partial / Total
Cart	Contains	Product	Many-to-Many	Partial / Partial
Order	Includes	Order_Item	One-to-Many	Total / Total
Product	Appears In	Order_Item	One-to-Many	Partial / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Delivery	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total
Product	Has	Inventory	One-to-One	Total / Total